Fragmented Housing Regulation as a Fiscal-Federalism Breakdown

A new measure of discretionary review, and the leakage of state preemption

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The question, and what is new

California has spent a decade preempting local zoning (SB 9/10, the builder's remedy). Yet the lever localities actually pull — **discretionary review** — is left untouched and goes *unmeasured*.

The contribution. I build the first item-level dataset of the discretionary margin from planning-commission minutes, and use it to ask:

- Q2 **What is the leakage of preemption?** When the state forces open *one* regulatory margin, do jurisdictions reassert restriction through the margin it leaves untouched? *Does preemption loosen supply, or get clawed back?*
- Q1 **How large is the distortion from fragmentation?** The gap between decentralized, non-internalized regulation and a regional planner who internalizes the spillover — *how much housing is "missing," at what welfare cost?*

Q2 is reduced-form and rests on the new data; Q1 layers a structural model on top. The order reflects what I am surest of.

The key object: restriction on two margins

A jurisdiction restricts supply by lowering the by-right envelope *or* raising the discretionary friction. The literature measures only the first.

By-right envelope $\bar{z}_j$

The zoning map: what a project obtains *ministerially*, no hearing.

- What LLM zoning measurement **sees** (Gupta & Barik; Bartik 2024).
- What preemption **targets**.

Discretionary tax τ_j (my data)

The review friction on projects exceeding $\bar{z}_j$: process exposure, mobilized opposition, approval-with-conditions, continuances, rare denial.

- What code-based measures **cannot** see.
- What preemption **leaves untouched**.

$$\text{Effective restriction} \quad r_j = g(\bar{z}_j, \tau_j), \qquad \text{lower } \bar{z}_j \textbf{ or } \text{raise } \tau_j.$$

Council sets the slow map $\bar{z}_j$; the planning commission sets the fast τ_j. **Their substitutability drives both questions** — and the binary approve/reject vote, which shows little variation in the minutes, misses it entirely.

Measuring τ_j: from minutes to data

Source. Planning-commission meeting minutes, parsed item-by-item via an LLM/NLP pipeline into a structured panel.

Extracted per agenda item

- Request type & project scale
- Stance-marked speaker counts (support / oppose)
- Outcome: approve, **approve-with-conditions**, continuance, denial
- # and type of conditions imposed
- Hearing exposure (continuances, re-hearings)

Why this is the friction, not the vote

Denials are rare; restriction shows up as *conditions, delay, and exposure* that deter applications before they reach a vote. τ_j aggregates these into a jurisdiction-level discretionary cost.

Validation: hand-coded subsample for precision/recall; cross-check τ_j against permit timelines and approval lags where available.

(Backup: one fully coded meeting item, raw text $\rightarrow$ extracted fields.)

Preemption and its leakage (Q2)

Preemption (ministerial mandates, by-right upzoning, the builder's remedy) raises the cost of a tight *envelope* $\bar{z}_j$ — but not of the *discretionary* tax τ_j.

Leakage

With $r_j = g(\bar{z}_j, \tau_j)$ and the two margins substitutes, a preemption that loosens $\bar{z}_j$ induces a still-restriction-demanding jurisdiction to **raise** τ_j in compensation (more conditions, more opposition entertained, more continuances). Effective restriction falls by *less* than the envelope change implies; **the conditional approval rate need not move at all**.

$$\text{Leakage } \in [0, 1]: \quad 0 = \text{preemption fully effective}, \quad 1 = \text{fully clawed back.}$$

Identification. SB 9/10 and builder's-remedy episodes are shocks to $\bar{z}_j$; I trace the τ_j response in the minutes around each episode. *This is the cleanest test in the project — a real shock, a directly measured response.*

Why it is a federalism question: the spillover

Regulation is set by jurisdictions ($J \approx 109$ in the Bay Area) far **finer** than the housing submarkets ($S \ll J$) and the *one* regional labor market they sit in.

A worker earns the regional wage $w_m = w(N)$, $w' > 0$ (agglomeration). So:

- the **benefit** of access-providing housing is *regional* (submarket price relief + higher w_m);
- the **cost** of that density is *local* disamenity.

A jurisdiction bears the full local disamenity of an approval but captures little price relief — most *leaks* to co-submarket neighbors. It under-supplies and **free-rides**.

Submarket price (the externality)

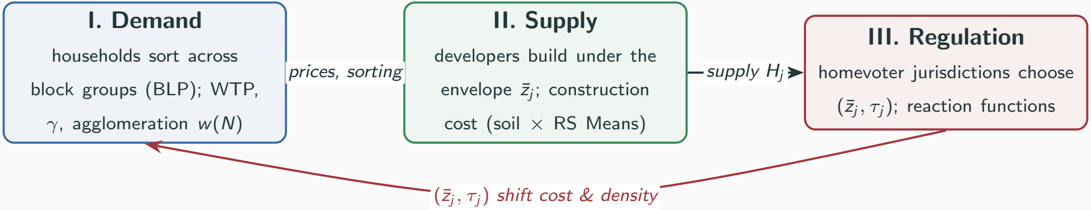
$$p_s = \frac{\alpha N_s}{H_s}, \qquad H_s = \sum_{j:\, s(j)=s} H_j$$

Housing built in *any* $j \in s$ lowers the price faced by incumbents in *every* $j \in s$.

The Oates wedge (Q1)

Decentralized Nash regulation is *tighter* than the internalized optimum — and the gap grows as j becomes atomistic within its submarket.

The framework: three layers, one equilibrium



- **Layers I–II** (demand & supply) recover the primitives needed to value any counterfactual allocation: who lives where, willingness to pay, how supply responds.
- **Layer III** (the *market for regulation*) is where the federalism breakdown lives, and where Q1 is posed.

The reaction function (Q1)

Because $p_{s(j)}$ depends on neighbors' regulation, Layer III is a **game**:

$$(\bar{z}_j, \tau_j) = R_j(\bar{z}_{-j}, \tau_{-j}), \qquad -j = \text{co-submarket neighbors only.}$$

- A neighbor *loosens* $\Rightarrow$ submarket price falls $\Rightarrow$ a homevoter-pivotal j *tightens*: regulation is **strategically substitutable**.

What the slope buys

Under the Decentralization Theorem's efficient case (no spillovers), jurisdictions need not respond to one another. A robustly *nonzero* slope, once confounders are ruled out, is the spillover made visible — its sign and magnitude quantify how badly the Oates condition fails.

$\Rightarrow$ requires a **multi-jurisdiction panel**; a single metro cannot, even in principle, reveal a reaction function.

Identification: the open problem I most want your read on

We see only *equilibrium* regulation, not the objective that generated it. Recovering $(\phi_j\lambda_j,\ g(\cdot),\ \text{demand} + w(N))$ is what licenses the counterfactuals.

The threat to R_j: a nonzero slope is consistent with

- the **spillover** (the mechanism), *but also*
- yardstick competition, or common submarket shocks (Brueckner 1998, 2003).

Separating these is the crux — and where I would most value your guidance.

Leverage I have:

- **Soil cost-shifter** (SSURGO foundation cost × RS Means), residualized against seismic / slope / open-space capitalization.
- **Preemption episodes** (SB 9/10, builder's remedy) as shocks to $\bar{z}_j$ that trace the τ_j response.
- **Submarket-size variation** in the free-riding wedge ($\partial p_s / \partial \text{supply}_j = -p_s / H_s$); BLP instruments for prices.

Routing: soil/topography is standard supply-side identification (Saiz). The strategic-interaction step is structural IO.

Data — demand side built, regulation side is the contribution

Regulation (Layer III) — novel

- **Discretionary margin τ_j:** item-level dataset from planning-commission minutes via LLM/NLP. Invisible to code-based measures.
- **By-right margin $\bar{z}_j$:** Gupta & Barik / Bartik LLM zoning — *your pointer*. I lean on theirs for $\bar{z}_j$; my data is the τ_j margin they can't see.

Demand & supply (Layers I–II) — assembled

- CoreLogic/Cotality **5.3M transactions** (price spine, geocoded to block group).
- ACS PUMS+tables, **LODES** job access ($w(N)$), TIGER + BG↔jurisdiction crosswalk.
- Amenities (open space, transit, schools), **crime** (SF/Oakland incident + FBI county), IRS migration.
- **SSURGO soil** cost-shifter instrument (BG-level).

~1.5 GB, fully documented (16-pp provenance + diagnostics report).

Where I am

Surest: the τ_j data and the leakage test (Q2) — new measure, real preemption shocks, directly observed response.

Most uncertain: identifying the reaction-function slope (Q1) against yardstick competition and common shocks; and the soil-cost exclusion restriction.

1. Does the leakage result stand on its own as a first paper, with Q1 as the structural extension — or would you sequence it differently?
2. On R_j: is there variation you would trust to separate spillover from common shocks?
3. Is the soil cost-shifter excludable from sorting after the controls, in your view?
4. Practically — what would you do first?